

Control Systems 1

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Welcome!

Polybox



PW: jschul

Website

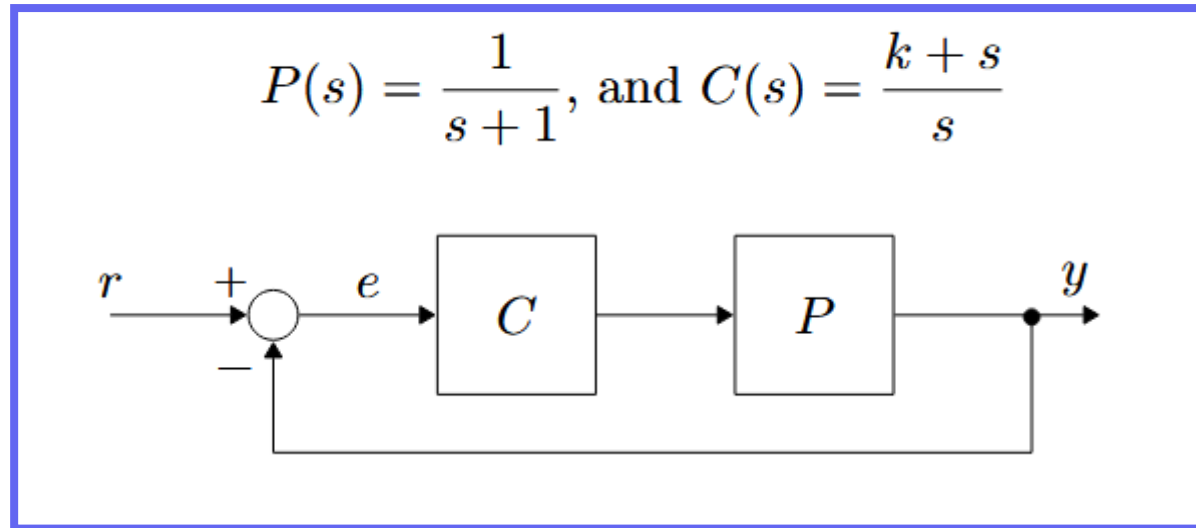


jschultev.github.io/personal_website

Today

- Repetition Session 11
- Theory Recap
 - Time Delays
 - Real World Compensators
 - Nonlinear Systems
- Q&A Session / Done

Repetition Session 11



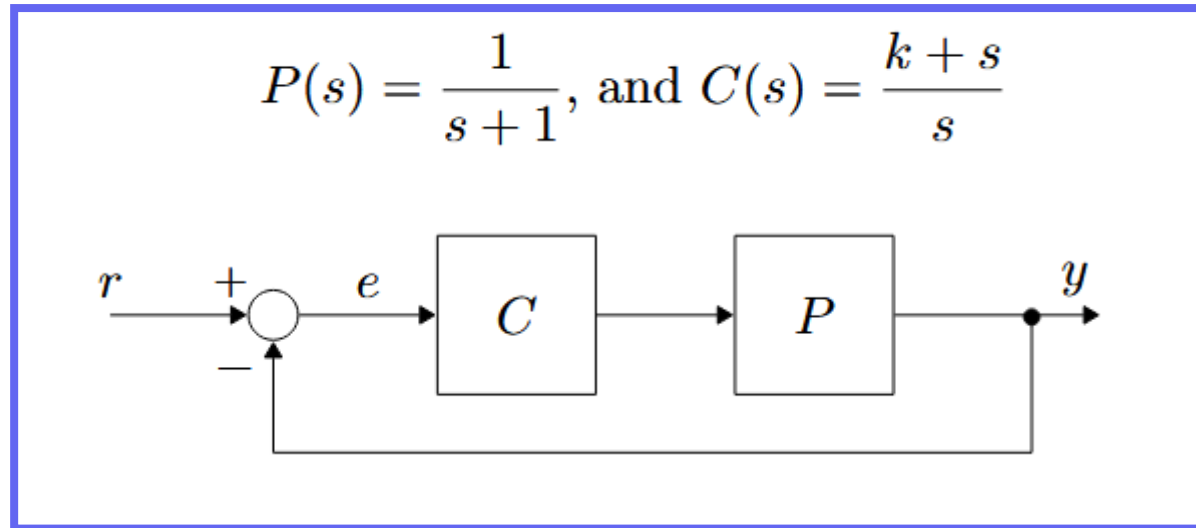
What k makes the gain crossover frequency be at $\omega_{gc} = 10$ rad/s ?

A) $400\pi^2$

C) $\frac{\pi}{20}$

B) 100

D) 0.25



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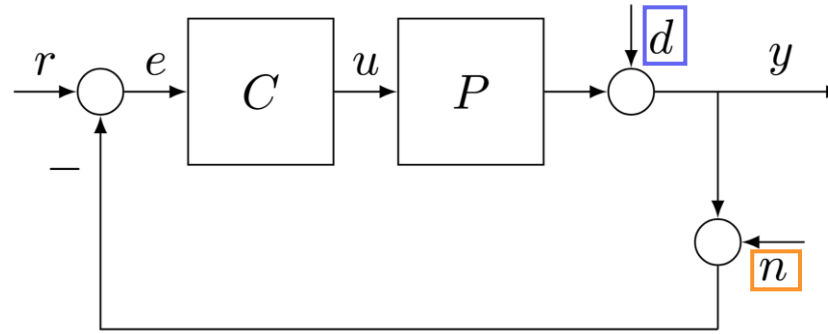
D) 0.25

$$P(s)C(s) = \frac{1}{s+1} \cdot \frac{k+s}{s} = \frac{k+s}{s^2+s}$$

$$L(j\omega_c) = \frac{k+10j}{-100+10j}$$

$$|A/B| = 1 \implies |A| = |B|$$

Closed Loop Transfer Function



Open Loop TF

$$L(s) = C(s)P(s)$$

maps: $e \rightarrow y$

Closed Loop TF
(Complementary Sensitivity)

$$T(s) = \frac{L(s)}{1+L(s)}$$

maps: $r \rightarrow y$, $n \rightarrow y$

Sensitivity Function

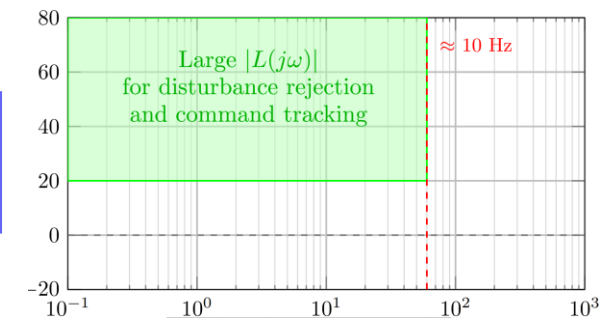
$$S(s) = \frac{1}{1+L(s)}$$

maps: $r \rightarrow e$, $d \rightarrow y$

Specifications

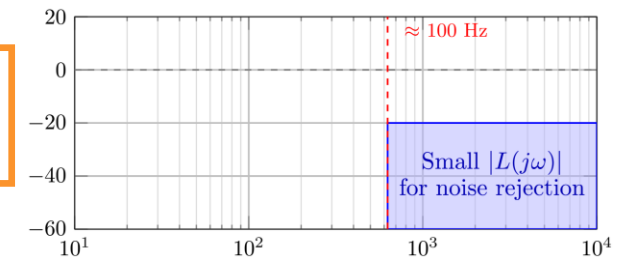
Since $S(s)$ maps the **disturbance signal $d(t)$ to the output $y(t)$** , we want $S(s)$ to be small for frequencies in which disturbances are present, therefore for low frequencies.

$$|S(j\omega)| \ll 1 \quad \text{for low } \omega \quad |S(j\omega)| = \left| \frac{1}{1+L(j\omega)} \right| \Rightarrow |L(j\omega)| \gg 1$$



Since $T(s)$ maps the **noise signal $n(t)$ to the output $y(t)$** , we want $T(s)$ to be small for frequencies in which noise is present, therefore for high frequencies.

$$|T(j\omega)| \ll 1 \quad \text{for high } \omega \quad |T(s)| = \left| \frac{L(s)}{1+L(s)} \right| \Rightarrow |L(j\omega)| \ll 1$$



↓

$$|T(s)| \approx |L(s)|$$

Tradeoff (Waterbed Effect)

$$S(s) + T(s) = \frac{1}{1 + L(s)} + \frac{L(s)}{1 + L(s)} = \frac{1 + L(s)}{1 + L(s)} = 1$$

We can see, that when summing the two functions, **they add up to 1**, meaning when one is small, the other one is big. So in the end we have a trade off at a given frequency between these two.

However, an important observation that helps us solve this dilemma, is that **disturbances** are present at **low frequencies** (up to 10Hz), and **noise** is present at **high frequencies** (more than 100Hz).

$$|L(j\omega)| \gg 1 \quad \text{for low } \omega,$$

$$|L(j\omega)| \ll 1 \quad \text{for high } \omega$$

Bandwidth

Hint: Often the Bandwidth is asked in Hz , then use $f = \frac{\omega}{2\pi}$

The **Bandwidth (BW)** of a system tells us the maximum frequency for which our system can track a command with a factor of ≈ 0.7 , which corresponds to

$$|T(j\omega)| > \frac{1}{\sqrt{2}} \quad (\approx -3\text{dB})$$

It can be interpreted as **how fast** the closed-loop can react to changing references.

Through some derivation we find, that $\omega_{\text{bw}} \approx \omega_c$

Meaning, our open loop crossover frequency is equal to our closed loop bandwidth

For this assumption to hold, we will always want a slope of -20dB/dec at the crossover frequency

Synthesis - Brute Force Approach

1. We start by designing an appropriate open loop TF $L_{des}(s)$ that fulfills all our desires.
2. We remember that we can write $L_{des}(s) = C(s)P(s)$
3. Now we can find our controller C with $C(s) = \frac{L_{des}(s)}{P(s)}$ where P is our plant.

This might seem easy to do, but:

- The open loop must be stable and minimum-phase. If not, **instabilities will just get cancelled**
- The compensator / controller may be **unnecessarily complicated**
- It may be that the compensator is **not causal**

Synthesis - Loop Shaping Approach

1. Rely on standard control elements with known characteristics and merge them

$$C(s) = C_1(s) \cdot C_2(s) \cdot \dots \cdot C_n(s)$$

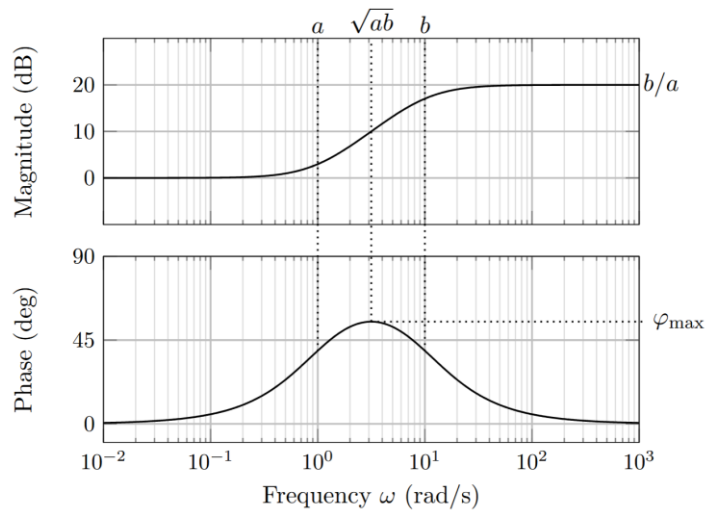
The most basic (but super useful) ones are:

Known	{	Proportional gain: k ,
		Integrator: $\frac{1}{s}$,
New	{	Lead element: $\frac{s/a+1}{s/b+1}$, with $0 < a < b$
		Lag element: $\frac{s/a+1}{s/b+1}$, with $0 < b < a$.

Lead

$$C_{\text{lead}}(s) = \frac{s/a + 1}{s/b + 1}, \quad 0 < a < b$$

Minimum Phase Zero before Stable Pole



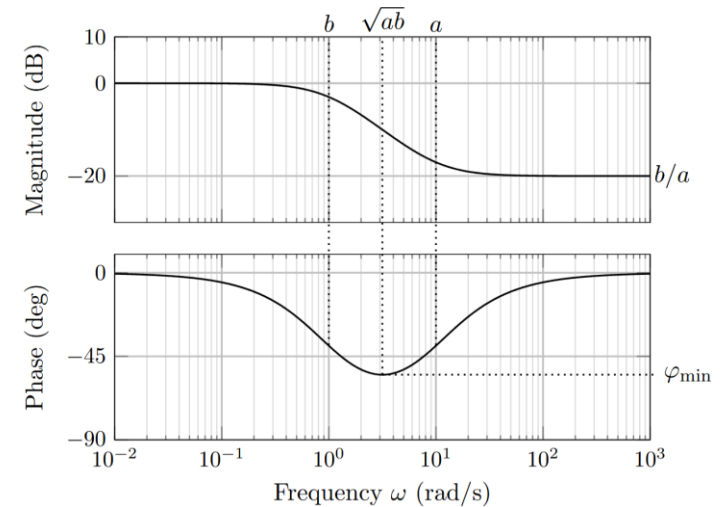
- Magnitude increase of $\frac{b}{a}$
- +20dB/dec between a and b
- Maximum phase **increase**

$$\varphi_{\max} = 2 \arctan \left(\sqrt{\frac{b}{a}} \right) - 90^\circ$$

Lag

$$C_{\text{lag}}(s) = \frac{s/a + 1}{s/b + 1}, \quad 0 < b < a$$

Stable Pole before Minimum Phase Zero



- Magnitude decrease of $\frac{b}{a}$
- -20dB/dec between a and b
- Maximum phase **decrease**

From / For ZF

Lead Compensators

A lead compensator is a loop element that increases the phase margin. However, it also makes a system more sensitive to noise.

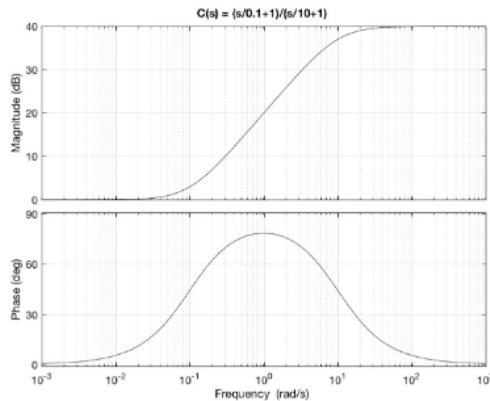
$$C_{\text{lag}} = \frac{s/a + 1}{s/b + 1} = \frac{b}{a} \cdot \frac{s + a}{s + b}, \quad 0 < a < b$$

$|L(j\omega)|$ changes by +20dB/dec between a and b . The maximum increase of φ is:

$$\Delta\varphi \approx 2 \cdot \arctan(\sqrt{b/a}) - 90^\circ$$

Design procedure:

1. Pick the desired crossover frequency $\omega_c = \sqrt{ab}$
2. Pick b/a depending on the desired phase increase
3. Possibly add a proportional gain k to set ω_c back to the desired frequency ($|kL(j\omega)| = 1$)



Lag Compensators

A lag compensator is a loop element that improves disturbance rejection and command tracking by decreasing the sensitivity to noise. However, it also decreases the phase margin of the system and can hence make it unstable:

$$C_{\text{lag}} = \frac{s/a + 1}{s/b + 1} = \frac{b}{a} \cdot \frac{s + a}{s + b}, \quad 0 < b < a$$

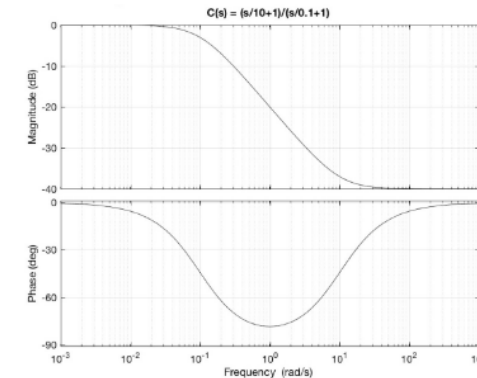
$|L(j\omega)|$ changes by -20dB/dec between a and b . The maximum decrease of φ is:

$$\Delta\varphi \approx 2 \cdot \arctan(\sqrt{b/a}) - 90^\circ$$

Careful: As the phase decreases, $\Delta\varphi$ is negative.

Design procedure:

1. Choose a/b as the desired increase in magnitude at low ω .
2. Pick $\sqrt{ab}$ as far as possible from the desired crossover frequency ω_c to not risk instability.



Theory Recap

Time Delays

Time Delays...?

In the previous weeks we saw how to properly design a controller for a given dynamical system. These controllers are often realised through computers and programs.

These two however, have a **finite computation time**, which means that the control input to a certain error / reference has some **delay**.

Think about driving a car: There will always be a slight time delay between applying the gas pedal and the car accelerating. This differs from car to car, but will always be present due to physical reasons.

From now on we will handle **Time Delays as operators** and analyze how they act on a system.

Time Delay

Time Delays will shift the input signal $u(t)$ by $T \geq 0$ units of time, resulting in the output:

$$y(t) = u(t - T)$$

The Time-Delay operator is **Linear** and **Time-Invariant**.
It therefore perfectly fits into our LTI systems!

To derive the Time-Delay operator and TF we will as usual plug in $u = e^{st}$, which will lead us to:

$$y(t) = e^{s(t-T)} = e^{st} e^{-sT} = e^{-sT} u(t)$$

The TF is therefore given by e^{-sT}

Time Delay Operator

The operator e^{-sT} may be LTI, but unfortunately it is **not rational**.
Therefore we cannot rely on classical tools such as the Root Locus method.

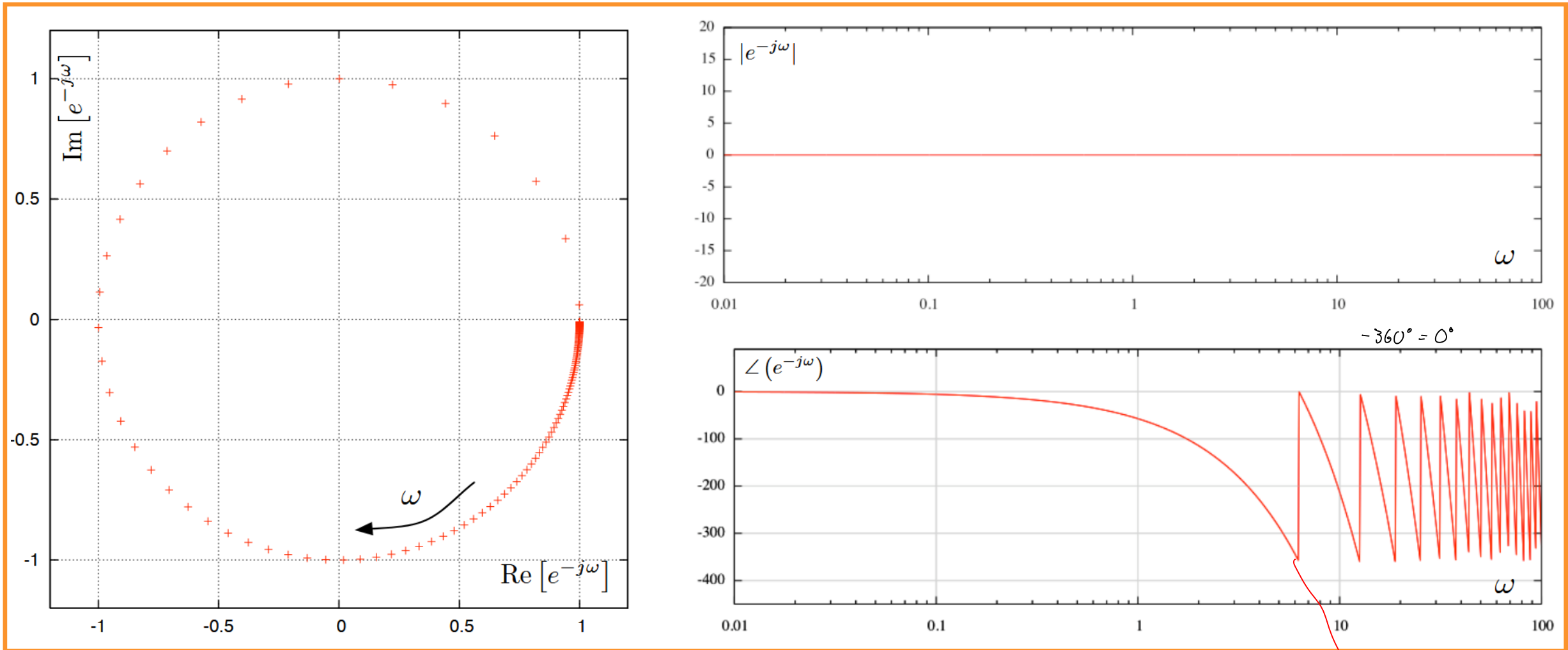
But let's for now look at the frequency response. Plugging in $s = j\omega$ we get:

$$\boxed{|e^{-j\omega T}| = 1} \quad \boxed{\angle(e^{-j\omega T}) = -\omega T}$$

From this we can deduce 2 important consequences:

- The **Magnitude is constant** and equal to one for all frequencies
- The **phase decreases linearly** with ω , meaning:
a sinusoidal signal gets more and more shifted

Time Delay Plot



Time Delay Consequences

$$L'(j\omega) = e^{-j\omega T} L(j\omega)$$

We will now compare L' with L .

Looking at the phase margin, for the system without time delay we find $\varphi_{m,0} = 180^\circ + \angle L(j\omega_c)$

Crossover frequency



For the time delay system:

$$\varphi_{m,T} = 180^\circ + \angle L'(j\omega_c) = 180^\circ + \angle L(j\omega_c) - \omega_c T = \varphi_{m,0} - \omega_c T$$

- For a fixed delay T , **increasing the crossover frequency ω_c reduces the phase margin.**
- For a fixed crossover frequency, **increasing the delay T reduces the phase margin.**

Controller Design with Time Delays

1. Design a feedback control system **ignoring** the time delay.
2. Check the effective phase margin (with time delay). If too small:
 - Increase phase at crossover (e.g. lead compensator)
 - Decrease crossover frequency (e.g. reduce the gain)
3. Iterate until satisfied

Unrelated Meme Break



Padé Approximation

As already mentioned, e^{-sT} is not a rational function. But we would still like to do root locus! Therefore, we will approximate this exponential time delay as a ratio of two polynomials (a.k.a. Padé approximation)

$$e^{-sT} \approx k \frac{s + p}{s + q}$$

To get the coefficients we will compare it to the Taylor series expansion: $k \frac{s + p}{s + q} = 1 - sT + \frac{1}{2}(sT)^2$

Finally we find:

$$e^{-sT} \approx \frac{\frac{2}{T} - s}{\frac{2}{T} + s}$$

Which means that we can now handle the our TF just as at the beginning, using tools like root locus.

- ☐ A Time delays are linear systems/operators.
- ☐ B Let $L(s)$ represent the transfer function of a linear time-invariant system without delays. Let $\tilde{L}(s)$ represent the transfer function of the same system as $L(s)$, where the output of the system now is delayed by 5s, i.e. $\tilde{L}(s)$ is a delayed version of the system represented by $L(s)$. It holds that, $\forall \omega \in \mathbb{R}_{\geq 0}$,

$$\begin{aligned} |\tilde{L}(j\omega)| &= |L(j\omega)|, \text{ and,} \\ \angle \tilde{L}(j\omega) &= \angle L(j\omega) - 5\omega. \end{aligned}$$

- ☐ C The transfer function of a time delay is a rational transfer function.
- ☐ D Time delays can cause closed-loop systems to become unstable.
- ☐ E Time delays can cause significant performance limitations to the achievable bandwidth of closed-loop systems, i.e. the maximal bandwidth while retaining closed-loop stability. More specifically, when a time delay is introduced to a closed-loop system, the achievable bandwidth decreases.

A)

B)

C)

D)

E)

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**Some Add-Ons since we have enough time!
But important!**

Real World Compensators

Compensators

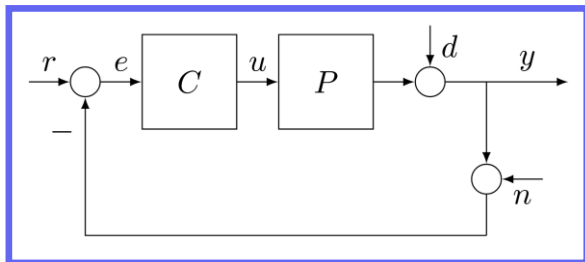
We learned, that compensators usually take the form of a TF, which we can always write as:

$$C(s) = k + \frac{c_{n-1}s^{n-1} + c_{n-2}s^{n-2} + \dots + c_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_0}$$

Now how can we incorporate this math expression into a real physical device such as a computer or microcontroller (like an Arduino), running a program?

→ We want to go back in the **State Space**:

Note that the "input" to this system is the error signal $e = r - y$, the output is the control input u .



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ & & \ddots & & & \\ 0 & 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & \dots & -a_{n-1} & & \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$
$$C = [c_0 \quad c_1 \quad \dots \quad c_{n-1}], \quad D = [d]$$

Compensator Pseudo Code

Initialize the state to, e.g., $x \leftarrow 0$.

Let dt be some small time interval.

repeat (every dt seconds)

Read the reference value r .

Read the measured output value y .

Compute the error e .

Update the state as $x \leftarrow x + (Ax + Be) * dt$.

Compute the output as $u = Cx + De$.

Send the command u to the actuators.

until Forever

Nonlinear Systems

Nonlinear Systems (Basically a recap:)

Most real world systems are nonlinear systems. We however have only looked at linear systems and developed our tools for them. To mitigate this problem, we **linearized our system around an equilibrium point**:

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t))\end{aligned}$$



$$A = \left. \frac{\partial f(x, u)}{\partial x} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \bigg|_{(x_e, u_e)}$$

$$B = \left. \frac{\partial f(x, u)}{\partial u} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \vdots \\ \frac{\partial f_n}{\partial u} \end{bmatrix} \bigg|_{(x_e, u_e)}$$

$$C = \left. \frac{\partial h(x, u)}{\partial x} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial h}{\partial x_1} & \cdots & \frac{\partial h}{\partial x_n} \end{bmatrix} \bigg|_{(x_e, u_e)}$$

$$D = \left. \frac{\partial h(x, u)}{\partial u} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial h}{\partial u} \end{bmatrix} \bigg|_{(x_e, u_e)}$$

The **Hartman-Großman** theorem states that if the **linearized system is closed-loop BIBO stable**, then the nonlinear system is also stable in a range about the equilibrium. We just don't know how large that region is

Why do I tell you this

This is just for you to see the entire scope of controller designing.

When designing for a nonlinear system follow these steps:

1. Design a linear compensator for the linearized model.
2. If the closed-loop system is stable, the nonlinear system will also be stable in a neighborhood of the equilibrium.
3. Check in a nonlinear simulation whether your design is robust to typical deviations.

HS 2023 (Some Repetition)

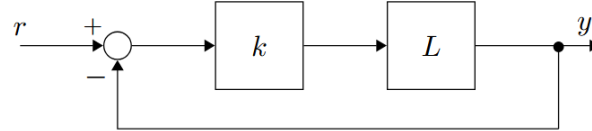


Figure 22: Closed-loop system T . Linear time-invariant system L and proportional gain k .

Q38 (1 Points) Mark all correct statements.

- ☐ A For any linear time-invariant system L , by relying on the Nyquist criterion, we can assess the stability of the closed-loop system T as a function of k .
- ☐ B For rational transfer functions $L(s)$, the root locus of $L(s)$ shows the poles of the closed-loop system T in the complex plane as a function of k .
- ☐ C For non-minimum phase and unstable linear time-invariant systems L , by relying on the Bode plot of L , we can reliably assess the stability of the closed-loop system T by checking whether $|L(j\omega_{pc})| < 1$, where ω_{pc} is such that $\angle L(j\omega_{pc}) = -180^\circ$.
- ☐ D Both the Polar plot of $L(s)$ and the Bode plot of $L(s)$ are representations of the complex number $L(j\omega)$ for $\omega \geq 0$.

A)

C)

B)

D)

HS 2023 (Some Repetition)

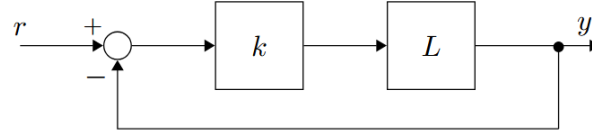


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Bode only for mp and stable!

A)

C)

B)

D)

Q&A Session / Done

Feedback



jschultev.github.io/personal_website/Feedback